

Brazil's soy land footprint is shifting toward China and toward the agricultural frontier

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Abstract

TODO: Results-only working document. The full abstract lives in the main manuscript. Placeholder so the file compiles standalone.

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1 Results

1.1 Two decades of footprint growth and a hand-off between consumers

The land-use footprint of Brazilian soy embodied in consumption grew from 9.4 Mha in 2001 to 41.0 Mha in 2022, closely tracking the expansion of the harvested area itself (Fig. 1a). The defining structural change of the period is a hand-off between the two dominant consumers (Fig. 1b). In 2001 the EU-27 accounted for 54% of the footprint and China for 11%; the two curves cross in 2009–2010, and by 2022 China alone accounts for 42% (17.2 Mha) against an EU-27 share below 20% (7.8 Mha). The EU footprint is remarkably stable in absolute terms (5.1 Mha in 2001, 7.8 Mha in 2022): virtually all of the growth in Brazilian soy land over two decades served demand growth in China and the rest of Asia (1.3 to 12.0 Mha), not in Europe.

1.2 From commodity exports to embodied animal products

Throughout the period, most of the footprint reaches consumers in commodity form, as beans, soybean meal, or oil (Fig. 2). From 2010 onwards, when the end-use attribution

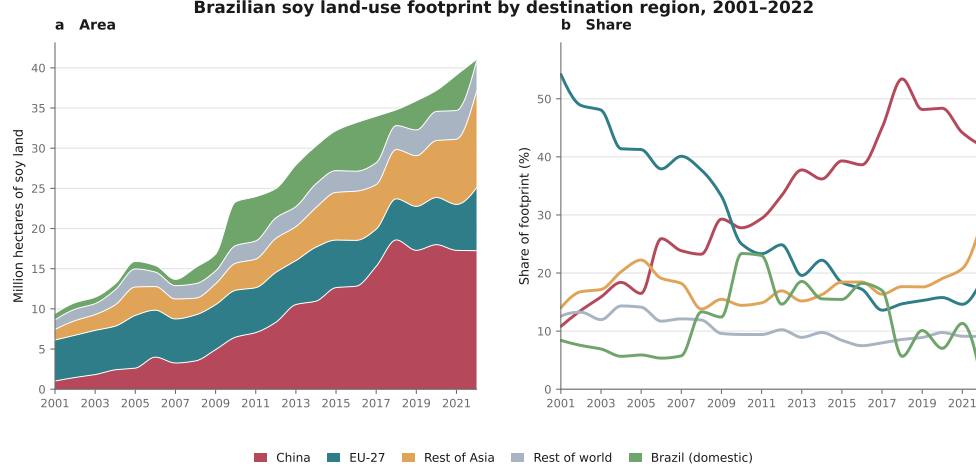


Fig. 1 Brazilian soy land-use footprint by destination region, 2001–2022. (a) Absolute footprint (million hectares of soy land embodied in consumption). (b) Regional shares of the total footprint. Years before 2010 are computed on the FABIO v1.1 backend and 2010 onwards on FABIO v2 (Methods); the vintage seam shifts the Brazil-domestic share by a few percentage points around 2009/2010 but does not affect the ranking or the crossing point.

is fully comparable across years, roughly a quarter of the footprint is embodied in animal products (meat, dairy and eggs) consumed at the destination, and the non-food and industrial share roughly doubles after Brazil’s 2008 biodiesel blending mandate: the non-food footprint rises from 0.8 Mha in 2007 to 1.5 Mha in 2008 and 2.5 Mha by 2013. Shares before 2010 are shown for completeness but are not directly comparable, because the v1.1 balance sheets book most soy-chain consumption at the commodity level (Methods).

1.3 The footprint marches northwest

Spatially, the two decades redraw the map of where consumed soy land sits (Fig. 3). In 2001 the footprint concentrated in the southern states and an emerging Mato Grosso core. By 2022 the centre of mass has moved decisively into the Cerrado frontier: northern Mato Grosso and the MATOPIBA region dominate, and the municipal pattern of China’s footprint, nearly empty in 2001 (1.0 Mha), spans the entire frontier by 2022 (17.2 Mha). The EU-27 pattern, by contrast, changes far less, consistent with its flat absolute footprint.

1.4 Validation: agreement with Trase grows with flow size

We validate the modelled municipality-to-country export flows against the independent, shipment-based Trase supply-chain data (2004–2022). Pooling all 127,862 municipality $\times$ destination $\times$ year flows above one tonne, the log-flow correlation with Trase is $r = 0.68$ and 54% of flows fall within a factor of three of the Trase value. Agreement rises strongly with flow size (Fig. 4): small flows scatter widely, while the large flows that carry the footprint converge onto the 1:1 line, so the model is most

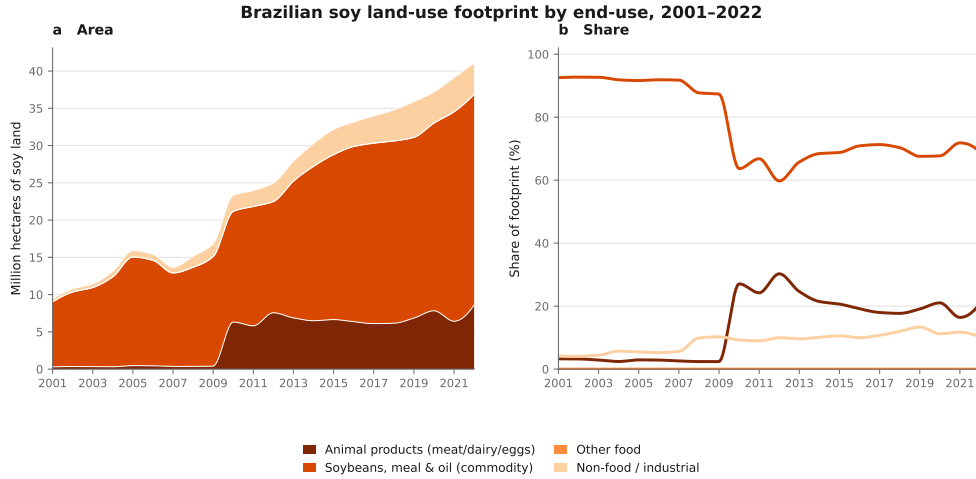


Fig. 2 Brazilian soy land-use footprint by end-use, 2001–2022. (a) Absolute footprint. (b) End-use shares. The step at 2009/2010 in the animal-products share is a classification seam between the FABIO v1.1 and v2 balance sheets, not a real dietary shift (Methods); end-use statements in the text are therefore restricted to 2010–2022.

reliable exactly where the volume is. This holds in every year, including the pre-2010 extension: the 2005 cross-section validates at $r = 0.65$, statistically indistinguishable from 2010 ($r = 0.73$). At the destination level, the pooled direct Pearson correlation with Trase rises from 0.43 in 2004 to around 0.7 from 2011 onwards, and our 2013 value (0.685) reproduces the published benchmark of the original single-year model (0.692). Per-tonne land intensities are essentially flat across destinations (about 0.33 ha per tonne of soybean-equivalent, the inverse of the national yield), with the EU-27 and China within $\pm 4\%$ of each other over 2010–2015.

Brazilian soy land-use footprint by destination, municipal origin, 2001 vs 2022
Hectares of soy land embodied in consumption (F_{mass}). All panels share one colour scale; sqrt-scaled.

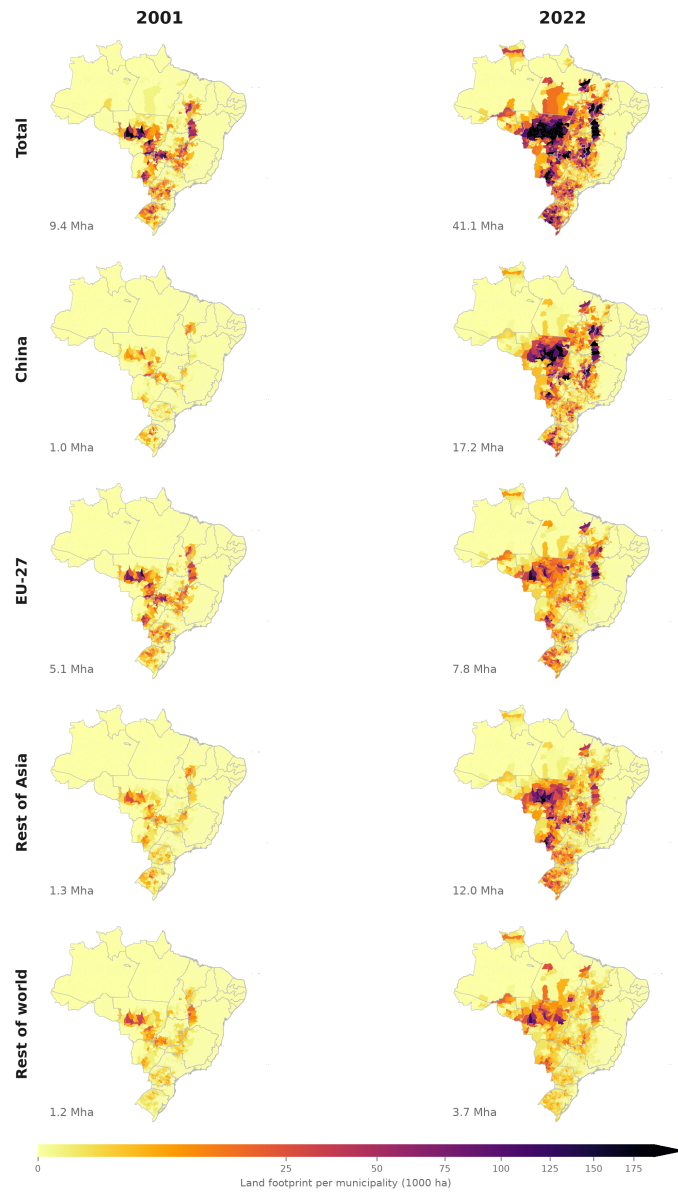


Fig. 3 Municipal origin of the Brazilian soy land-use footprint, 2001 versus 2022, for the total and the four major destination groups. All panels share one colour scale (square-root scaled).

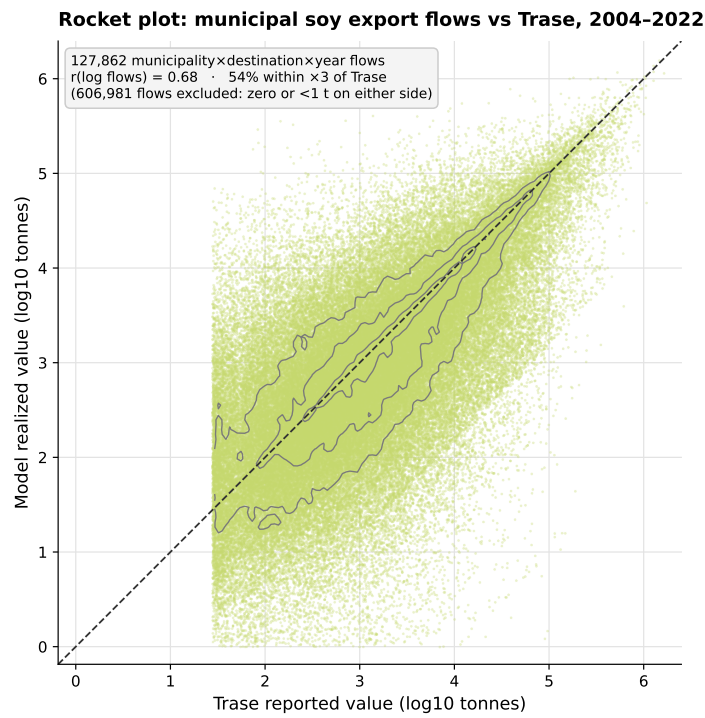


Fig. 4 Rocket plot: modelled municipal soy export flows against Trase, all years pooled (2004–2022), both axes log10 tonnes. Contours show point density; the dashed line is 1:1. Flows below one tonne on either side are excluded; the visible lower bound on the Trase axis (about 28 t) is the granularity floor of the Trase composite itself.